SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS MASTER IN CONTROL

1. Exercise

a) The function $f(x)$ that defines the system's equations is polynomial in both variables. Hence it is locally Lipschitz and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [2]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathscr{N}_1 = \{x : x_2 = x_1^3 - 2x_1\}.$$

This is the equation of a cubic which intersects the x_1 axis in $\pm\sqrt{2}$ and 0. Moreover, the slope in 0 of the nullcline is -2. All solutions above the cubic move eastwards and those below west. For the second nullcline we see:

$$\mathscr{N}_2 = \{x : x_1 = x_2^3 - 2x_2\}.$$

The second nullcline is therefore of the same shape as nullcline $\mathscr{N}_1$, but symmetric with respect to the diagonal. All solutions to the right of the cubic move south and those to the left move north. [3]

c) There are 8 regions delimited by the nullclines. The direction of the vector-field in each one of them is shown in Fig. 1.1. [3]

d) From the direction of the arrows we notice that all regions whose interior intersects the diagonal (dotted line) are forward invariant (while the rest are not). [2]

e) To find equilibria of the system we look at nullclines intersections. There are 5 intersections. Due to symmetry of the nullclines these are either along the diagonal, or on the anti-diagonal. Hence we can find them using the extra constraint $x_1 = \pm x_2$. On the diagonal we see that $x_1 = x_1^3 - 2x_1$ yields $x_1 = 0$ and $x_1 = \pm\sqrt{3}$. Thus three equilibria are in $[0,0]', [\sqrt{3},\sqrt{3}]', [-\sqrt{3},-\sqrt{3}]'$. Instead for $x_1 = -x_2$ we see that $-x_1 = x_1^3 - 2x_1$ yields $x_1 = \pm 1$, thus giving 2 more equilibria in $[1,-1]$, $[-1,1]$. [2]

f) To linearize the system we first compute the Jacobian matrix $\frac{\partial f}{\partial x}$ at a generic value of x.

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} -3x_1^2+2 & 1 \\ 1 & -3x_2^2+2 \end{bmatrix}.$$

At the equilibrium $x_{e1} = [0,0]'$ we see that:

$$\frac{\partial f}{\partial x}(x_{e1}) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_1(s) = s^2 - 4s + 3$, whose roots are 1 and 3. Two positive eigenvalues correspond to the phase potrait of an unstable node (hyperbolic). At the equilibria $x_{e2} = [\sqrt{3},\sqrt{3}]'$ and $x_{e3} = -[\sqrt{3},\sqrt{3}]'$ we have the Jacobian:

$$\frac{\partial f}{\partial x}(x_{e2}) = \begin{bmatrix} -7 & 1 \\ 1 & -7 \end{bmatrix}.$$

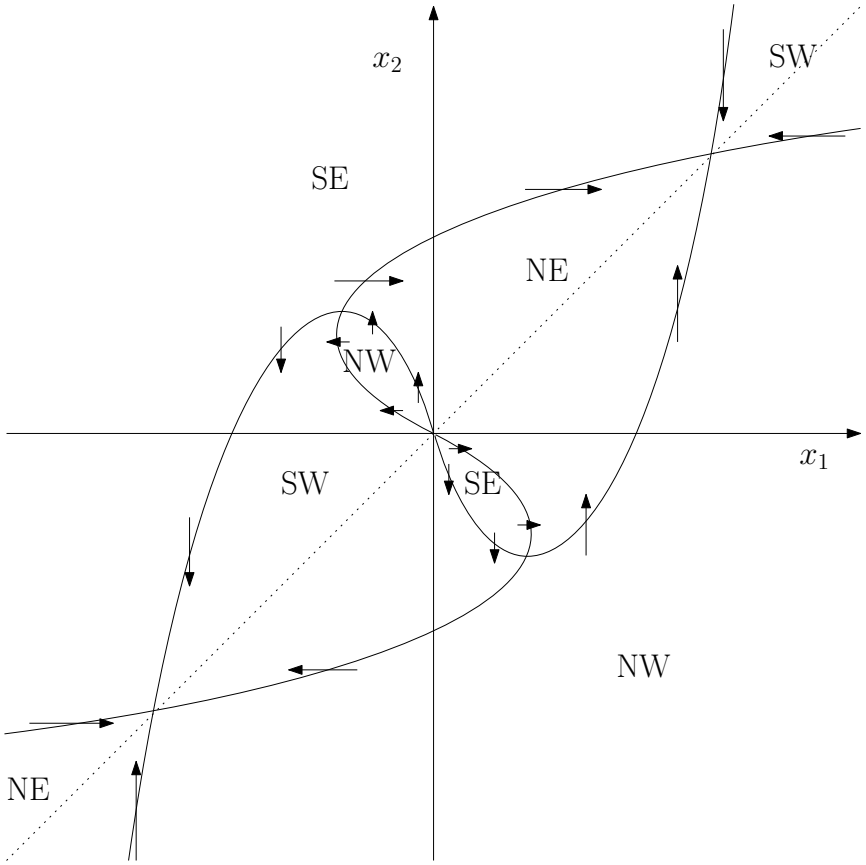


Figure 1.1 Nullclines and vector-field orientations

Its characteristic polynomial is $\chi_2(s) = \chi_3(s) = s^2 + 14s + 48$ with roots in -8 and -6. Two negative roots correspond to the phase portrait of a stable node (hyperbolic). At the equilibria $x_{e4} = [1,-1]'$ and $x_{e5} = [-1,1]'$ the Jacobian takes the following form:

$$\frac{\partial f}{\partial x}(x_{e4}) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_4(s) = \chi_5(s) = s^2 + 2s$ with eigenvalues in 0, -2. This is not hyperbolic. From the nullclines we see that the local phase portrait appears similar to that of a saddle point. [3]

g) To show that the diagonal is invariant it is enough to see that for $x_1 = x_2$, it holds $\dot{x}_1 = \dot{x}_2$. This is trivially seen given the symmetry of the equations. [3]

h) The global phase portrait is sketched in Fig. 1.2. [2]
The global phase-portrait is often something that students struggle with. Keep in mind that local phase portraits should merge while respecting the following constraints: 1) solutions should not cross each other, except at equilibria 2) nearby solutions should move parallel to each other with consistent speed orientation

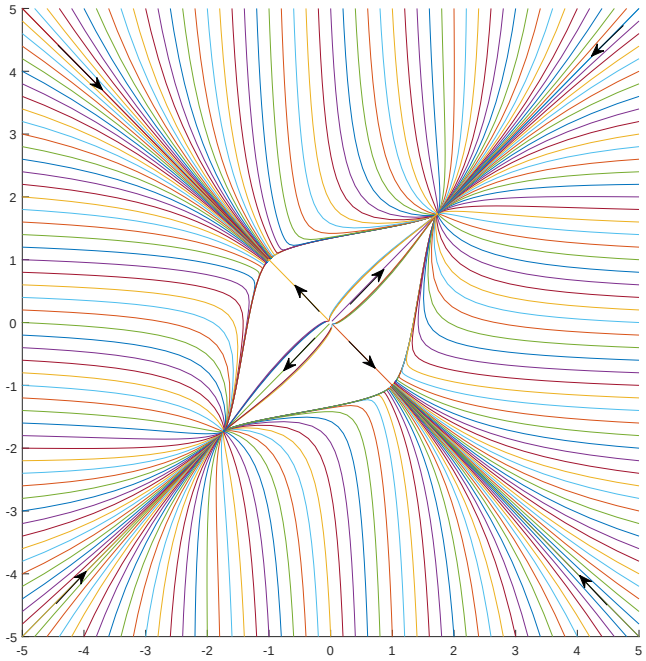


Figure 1.2 Phase portrait

2. Exercise

a) We may take P to be the identity. Indeed:

$$A_1' + A_1 = \begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix} < 0$$

and similarly:

$$A_2' + A_2 = \begin{bmatrix} -2 & 1 \\ 1 & -4 \end{bmatrix} < 0.$$

Two marks allocated for writing up systems of conditions on P and three marks for finding solution. [5]

This question has caused several problems to student. Notice that apparently for a symmetric matrix 3 unknown parameters need to be found. However, because the conditions are linear, effectively one of them (for instance p_{11}) can always be fixed to be 1. In this case in particular a diagonal matrix was enough, which really leaves only one parameter to decide.

b) To see that $x = 0$ is an equilibrium for all $d \in \mathbb{R}$ it is enough to compute:

$$f(0,d) = dA_1 \cdot 0 + (1-d)A_2 \cdot 0 = 0.$$

[1]

c) Next, we choose as a Lyapunov function $V(x) = x'x$. Its derivative along solutions of the system is:

$$\dot{V} = \dot{x}'x + x'\dot{x} = dx'(A_1' + A_1)x + (1-d)x'(A_2' + A_2)x$$

$$\leq d\bar{\lambda}(A_1' + A_1)x'x + (1-d)\bar{\lambda}(A_2' + A_2)x'x = (-3+\sqrt{2})x'x.$$

where $\bar{\lambda}(M)$ denotes the largest (real) eigenvalue of M. In particular, $\bar{\lambda}(A_2' + A_2) = \bar{\lambda}(A_1' + A_1) = -3 + \sqrt{2}$. [2]

Hence $V(x)$, positive definite and radially unbounded, has a negative definite derivative (uniformly for $d \in [0,1]$) and this shows that the origin is a UGAS equilibrium. [2]

d) To study the scalar system $\dot{z} = -4z^3 + z$ we first compute its equilibria:

$$-4z^3 + z = 0 \Rightarrow z \in \{\pm 1/2, 0\}.$$

Moreover, $-4z^3 + z < 0$ for $z > 1/2$, and $z \in (-1/2, 0)$, while $-4z^3 + z > 0$ for $z < -1/2$ and $z \in (0, 1/2)$. In particular then, exploiting monotonicity of solutions for scalar systems, we see that $\omega(z_0) = 0$ for $z_0 = 0$, $\omega(z_0) = \{1/2\}$ for $z_0 > 0$ and $\omega(z_0) = \{-1/2\}$ for $z_0 < 0$. [3]

e) The equations of the cascaded system read:

$$\begin{aligned} \dot{z} &= -4z^3 + z \\ \dot{x} &= (1+z)A_1 x/2 + (1-(1+z)/2)A_2 x. \end{aligned}$$

[2]

f) Notice that, when the state of system Σ_2 is at equilibria $-1/2, 0, 1/2$, the corresponding output values are $1/4, 2/4, 3/4$ (respectively). Therfore, the previous analysis implies that regardless of the initial condition z_0, there exists a finite time T such that $y(t) \in [0,1]$ for all $t \geq T$. (Notice that solutions of the cascaded

system are defined for all times since f is linear in x). Hence, regardless of z_0 and x_0 there exists a finite time T such that $d(t) \in [0,1]$ for all $t \geq T$. Because of UGAS, there exists β of class $\mathscr{KL}$ such that:

$$|x(t)| \leq \beta(|x(T)|, t-T) \qquad \forall t \geq T.$$

As a consequence, $x(t) \to 0$ as $t \to +\infty$. Overall the ω-limit set of the cascaded system is still a singleton for each initial condition. In particular, equal to $\{[-1/2,0,0]'\}, \{[0,0,0]\}$ and $\{[1/2,0,0]\}$ depending on the initial value for z_0, respectively negative, 0 or positive. Three marks allocated for correct answer and two for rigorous justification. [5]

3. Exercise

a) For a SISO control affine system,

$$\dot{x} = f(x) + g(x)u, \quad y = h(x),$$

passivity and losslessness amount to the following two conditions:

$$\frac{\partial S}{\partial x}(x) \cdot f(x) = 0$$
$$\frac{\partial S}{\partial x}(x) \cdot g(x) = h(x).$$

Hence we may require that:

$$\frac{\partial S}{\partial x_1}(x) 4x_2 - \frac{\partial S}{\partial x_2}(2x_1 + 2\sin(x_1)\cos(x_1)) = 0$$
$$\frac{\partial S}{\partial x_2}(x) = y = 4x_2.$$

[3]

Integrating the second one with respect to x_2 we see that $S(x)$ needs to be of the following form:

$$S(x) = 2x_2^2 + \tilde{S}(x_1),$$

for some $\tilde{S}$ function of x_1 only. Similarly, substituting $4x_2$ for $\frac{\partial S}{\partial x_2}$ in the first equation we see that we can take:

$$\frac{\partial S}{\partial x_1}(x) = 2x_1 + 2\sin(x_1)\cos(x_1).$$

Integration of the latter yields:

$$S(x) = x_1^2 + \sin^2(x_1) + \hat{S}(x_2),$$

for some suitable function $\hat{S}$ of x_2. Combining the previous expressions we see that:

$$S(x) = x_1^2 + \sin^2(x_1) + 2x_2^2$$

is a suitable storage function. [2]

b) We propose the following static output feedback:

$$u = -\frac{e^y - 1}{e^y + 1} := k(y).$$

Notice that this is an increasing function of y, such that $|k(y)| \leq 1$ for all $y \in \mathbb{R}$. Moreover $k(y) > 0$ iff $y < 0$ and $k(y) < 0$ iff $y > 0$. [2]

Input constraints: several students did not manage to fulfill constraints but were able to stabilize the system without the constraint. Notice that for a static SISO output feedback you can always saturate the feedback, while preserving passivity. This allows to essentially meet input constraints without any stability issues.

Overall, we let $V(x) := S(x)$ be a candidate Lyapunov function. We see that V is positive definite and radially unbounded, since $V(x) \geq x_1^2 + 2x_2^2$. For its derivative we have:

$$\dot{V} = [2x_1 + 2\sin(x_1)\cos(x_1), 4x_2][f(x) + g(x)u] = yu = yk(y) \leq 0.$$

Hence, $\dot{V}$ is negative semidefinite and $\text{Ker}[\dot{V}] = \{x : x_2 = 0\}$. Any invariant set contained in $\text{Ker}[\dot{V}] = \{x : x_2 = 0\}$ is therefore also contained in

$$\{x : x_2 = 0 \text{ and } -2x_1 - 2\sin(x_1)\cos(x_1) = 0\} = \{0\}.$$

Hence, the system is GAS by Lasalle's invariance principle. [3]

c) Considering the equation for $\dot{S}$ we see that:

$$\dot{S} = yu.$$

We propose the feedback:

$$u = y(S_{ref} - S(x)).$$

[2]

Consider next the Lyapunov function $W(x) := (S(x) - S_{ref})^2$. We see that:

$$\dot{W} = 2(S(x) - S_{ref})yu = -2y^2(S(x) - S_{ref})^2 \leq 0.$$

[3]

d) Since $\dot{W} \leq 0$ we have that $W(x_{cl}(t))$ is non-increasing along systems solutions. Moreover it is bounded below. Hence its limit as $t \to +\infty$ exists. [2]
Notice that the ω-limit set of solutions is non-empty and invariant and therefore is contained in a level set of W. This implies that solutions in $\omega(x_0)$ have zero $\dot{W}$. This implies zero y or zero $S(x) - S_{ref}$. The first case can be ruled out since y identically zero implies u identically zero and x_1 identically zero, which only occurs when $x_1(0) = x_2(0) = 0$ and $S_{ref} = 0$. [3]

4. Exercise

a) To compute the relative degree we consider the output and its Lie derivatives until the input u appears in the corresponding expressions:

$$\begin{aligned} y &= x_1 + x_2 \\ \dot{y} &= \dot{x}_1 + \dot{x}_2 = x_1 - x_2 \\ \ddot{y} &= \dot{x}_1 - \dot{x}_2 = 2\text{atan}(x_1) + 2(2+\cos(x_2))u. \end{aligned}$$

[2]

Since the input u only appears after 2 derivations, the relative degree is 2. Moreover, its coefficient $2(2+\cos(x_2))$ is non zero for all $x \in \mathbb{R}^3$ and therefore the relative degree is globally defined. [2]

b) We choose the feedback law:

$$u = \frac{v - 2\text{atan}(x_1)}{2(2+\cos(x_2))}.$$

With this choice the input-output equations read: $\ddot{y} = v$, which is indeed the equation of a double integrator. [3]

c) We use the following variables as a set of independent coordinates:

$$\xi_1 = y, \qquad \xi_2 = \dot{y}, \qquad z = x_3.$$

Accordingly we have the equations:

$$\begin{aligned} \dot{\xi}_1 &= \xi_2 \\ \dot{\xi}_2 &= v \\ \dot{z} &= -z^3 + \xi_1 \\ \\ y &= \xi_1 \end{aligned}$$

[3]

The above equations are in normal form. Moreover the equation for $\dot{z}$ describes the so called internal dynamics. [2]

d) The internal dynamics are one-dimensional, and fulfill the equation:

$$\dot{z} = -z^3 + \xi_1$$

where ξ_1 is regarded as the exogenous input signal. These equations define an ISS system. [2]
Indeed we can take the ISS Lyapunov function: $V(z) = z^2/2$ which is positive definite and radially unbounded. Along solutions of the system we see that:

$$\dot{V} = z(-z^3 + \xi_1) = -z^4 + z\xi_1 \leq -|z|^4 + |z||\xi_1|.$$

Assume $|z| \geq \chi(|\xi_1|) := \sqrt[3]{2|\xi_1|}$. Notice that χ is a class $\mathscr{K}_\infty$ function. We see that:

$$|z| \geq \chi(|\xi_1|) \Rightarrow |z|^3 \geq 2|\xi_1| \Rightarrow |z|^4/2 \geq |z||\xi_1| \Rightarrow \dot{V} \leq -|z|^4/2.$$

Hence V is an ISS Lyapunov function of type 2, and ISS of the internal dynamics hold. [2]

e) We may pick the stabilizing feedback as $v = -2\dot{y} - y$. [2]
This induces a cascade of a linear systems with two poles in -1 which is forcing the internal dynamics. Since the internal dynamics are ISS the cascaded interconnection is also Globally Asymptotically Stable. [2]